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TASK 4

INTRODUCTION

The process of deploying resources to a Kubernetes cluster using kubectl and exposing a service locally via Minikube. It includes commands to apply YAML configurations for deployments and services, followed by using the minikube service command to create a tunnel and access the service on the local machine.

vim t1.txt and vim t2.txt:

These commands open the vim text editor to create or edit files named t1.txt and t2.txt. These files likely contain YAML definitions for Kubernetes resources like deployments or services.

```
karunyamathivanan@LAPTOP-N670L8SV:~$ vim t1.txt
karunyamathivanan@LAPTOP-N670L8SV:~$ vim t2.txt
```

kubectl apply -f t1.txt:

This command applies the configuration defined in t1.txt to the Kubernetes cluster.

kubectl apply -f t2.txt:

Similar to the previous command, this applies the configuration in t2.txt to the cluster. If t2.txt defines a service, it will create or update the service.

```
karunyamathivanan@LAPTOP-N670L8SV:~$ kubectl apply -f t1.txt
deployment.apps/test created
karunyamathivanan@LAPTOP-N670L8SV:~$ kubectl apply -f t2.txt
service/test-service created
```

minikube service test-service:

This command starts a Minikube tunnel to expose the test-service to your local machine. The output shows the service's URL and target port. Minikube sets up this tunnel so you can access the service from your host machine.

```
karunyamathivanan@LAPTOP-N670L8SV:~$ minikube service test-service
-----
| NAMESPACE | NAME       | TARGET PORT | URL                               |
|-----|-----|-----|-----|
| default   | test-service | http/80      | http://192.168.49.2:30153 |
|-----|-----|-----|-----|
🚀 Starting tunnel for service test-service.
-----
| NAMESPACE | NAME       | TARGET PORT | URL                               |
|-----|-----|-----|-----|
| default   | test-service |             | http://127.0.0.1:38269 |
|-----|-----|-----|-----|
🌐 Opening service default/test-service in default browser...
👉 http://127.0.0.1:38269
! Because you are using a Docker driver on linux, the terminal needs to be open to run it.
^C 🛑 Stopping tunnel for service test-service.
```

Output Details:

- The **URL** field indicates the address to access the service.
- The first URL (e.g., 192.168.49.2:30153) is the internal Minikube cluster IP.
- The second URL (e.g., 127.0.0.1:38269) is the local machine's address where the service is exposed.

